

ChanningCGDS

Table of contents

Welcome	3
Docker image	4
RStudio Server	5
Session info	6
1 Road map	9
2 Computing platform concepts	10
3 Storage concepts for cohorts and experiments	11
4 Data structure concepts for genomic analysis	12
4.1 Self-descriptive data structures	12
4.1.1 Example: bulk RNA-seq analysis of smooth muscle treated with dexam- ethasone	12
4.1.2 Example: Zilionis Single-Cell RNA-seq in lung	12
5 Analytic software and workflows	18

Welcome

Package: ChanningCGDS **Authors:** Vincent Carey [aut, cre] **Compiled:** 2025-07-17 **Package version:** 0.98.2 **R version:** R version 4.5.1 (2025-06-13) **BioC version:** 3.22 **License:** MIT + file LICENSE

This is the landing page of the **BiocBook** entitled ChanningCGDS (CGDS = Computational Genomic Data Science).

This book introduces the reader to collaborative approaches to learning and teaching about research methods in the domains of whole genome sequencing, transcriptomics, epigenomics, and “spatial omics” with ultimate focus on lung diseases.

Docker image

A Docker image built from this repository (for linux amd64 platforms) is available here:

ghcr.io/vjcitn/channingcgds

Get started now

You can get access to all the packages used in this book in < 1 minute, using this command in a terminal:

Listing 0.1 bash

```
docker run -it ghcr.io/vjcitn/channingcgds:devel R
```

RStudio Server

An RStudio Server instance can be initiated from the **Docker** image as follows:

Listing 0.2 bash

```
docker run \  
  --volume <local_folder>:<destination_folder> \  
  -e PASSWORD=OHCA \  
  -p 8787:8787 \  
  ghcr.io/vjcitn/channingcgds:devel
```

The initiated RStudio Server instance will be available at <https://localhost:8787>.

Session info

 Click to expand

1 Road map

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2 Computing platform concepts

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3 Storage concepts for cohorts and experiments

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4 Data structure concepts for genomic analysis

4.1 Self-descriptive data structures

4.1.1 Example: bulk RNA-seq analysis of smooth muscle treated with dexamethasone

[rnaSeqGene workflow](#)

[pubmed link to original](#)

[study](#)

4.1.2 Example: Zilionis Single-Cell RNA-seq in lung

Immunity
Resource



Single-Cell Transcriptomics of Human and Mouse Lung Cancers Reveals Conserved Myeloid Populations across Individuals and Species

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<https://doi.org/10.1016/j.immuni.2019.03.009>

Figure 4.1: Title of paper

scRNAseq

```
library(scRNAseq)
library(SingleCellExperiment)
library(ggplot2)
```

```
zil = ZillionisLungData()
```

```
zil
## class: SingleCellExperiment
## dim: 41861 173954
## metadata(0):
## assays(1): counts
## rownames(41861): 5S_rRNA 5_8S_rRNA ... snosnr66 uc_338
## rowData names(0):
## colnames(173954): bcIIOD bcHTNA ... bcELDH bcFGGM
## colData names(16): Library Barcode ... used_in_NSCLC_non_immune
##   used_in_blood
## reducedDimNames(5): SPRING_NSCLC_all_cells
##   SPRING_NSCLC_and_blood_immune SPRING_NSCLC_immune
##   SPRING_NSCLC_non_immune SPRING_blood
## mainExpName: NULL
## altExpNames(0):
```

4.1.2.1 Why is this “self-descriptive”?

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4.1.2.2 Excursus: classes and methods

[SingleCellExperiment](#)

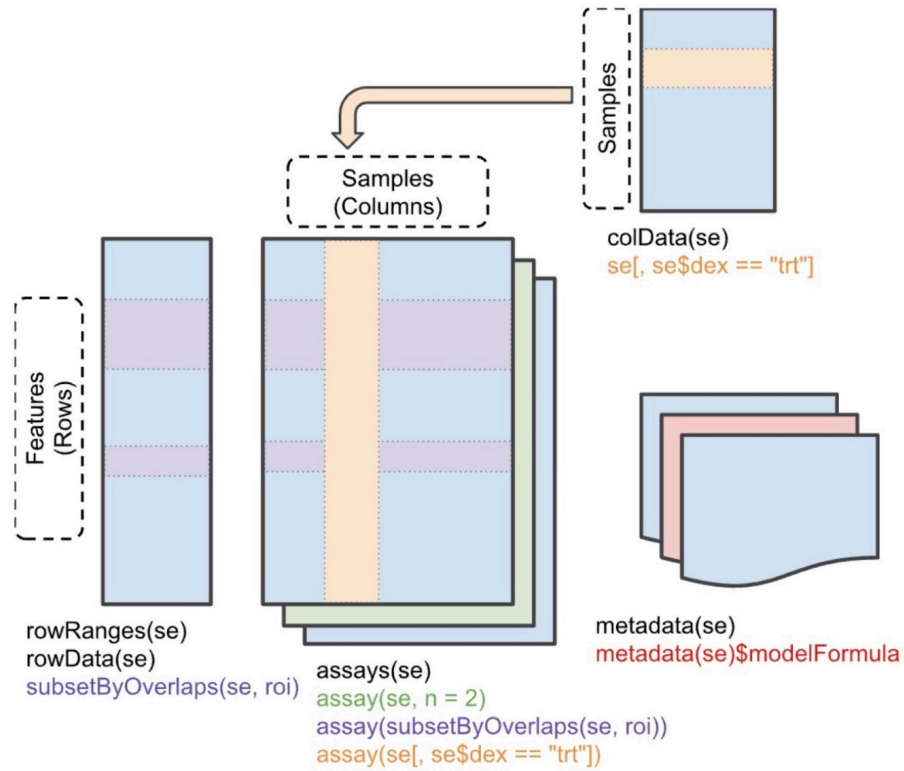


Figure 4.2: SummarizedExperiment schematic

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```
class(zil)
## [1] "SingleCellExperiment"
## attr("package")
```

```
## [1] "SingleCellExperiment"
is(zil, "SummarizedExperiment")
## [1] TRUE
```

“S4 object-oriented programming”

```
allm = methods(class="SummarizedExperiment")
grep("assay,", allm, value=TRUE)
## [1] "assay, SummarizedExperiment, character-method"
## [2] "assay, SummarizedExperiment, missing-method"
## [3] "assay, SummarizedExperiment, numeric-method"
```

4.1.2.2.1 How does this represent useful binding of metadata to data?

```
table(zil$Patient, zil$Tissue)
##
##      blood tumor
## p1 12763 21243
## p2 15826 14873
## p3  8165 30794
## p4  4906 11907
## p5     0 14049
## p6  3964 11300
## p7  6981 17183
```

```
library(ggplot2)
rd = as.data.frame(reducedDims(zil)[[3]])
rd = cbind(rd, type = zil$`Major cell type`)
rdp = na.omit(rd) # many cells not used
ggplot(rdp, aes(x=x, y=y, colour=type)) + geom_point(size=.5)
```

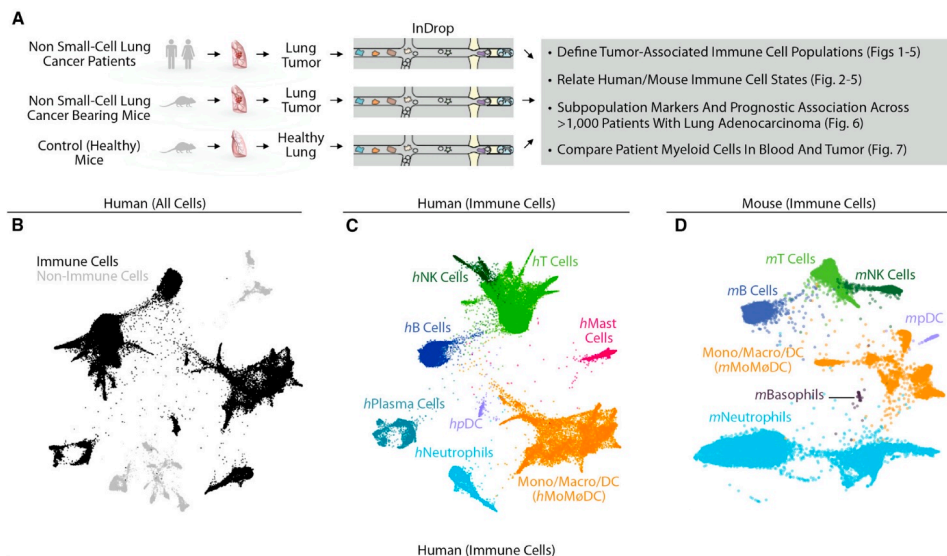
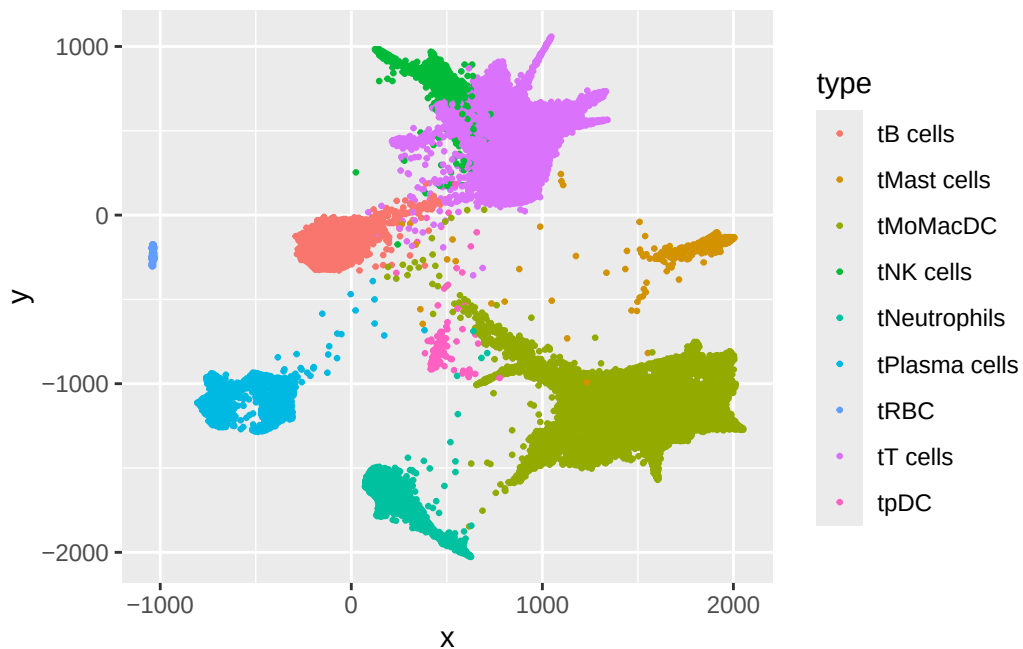


Figure 4.3: Fig 1 excerpt

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5 Analytic software and workflows

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